

Mohamed Shereef CH

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SUMMARY

Recent B.Tech graduate in Information Technology with experience designing and shipping full-stack applications, backend APIs, real-time systems, and data pipelines. Writes clean, maintainable code across Python and JavaScript; comfortable with databases, system design, third-party integrations, and cloud deployment. Has built and launched production products from scratch and contributed to open source.

EDUCATION

- Bachelor of Technology in Information Technology** 2026
MEA Engineering College, Kerala

TECHNICAL SKILLS

- Languages:** Python, JavaScript, TypeScript, SQL, Java, C
- Backend:** FastAPI, Node.js, REST API design, async/await, JWT & OAuth2, middleware, request validation (Pydantic), Supabase Edge Functions
- Frontend:** React, Vite, Tailwind CSS, shadcn/ui
- Databases:** PostgreSQL (schema design, joins, indexing, Row-Level Security), Supabase, Firebase Firestore, ChromaDB
- Real-Time & Systems:** Supabase Realtime, Ably Pub/Sub, Agora Voice SDK, WebSocket event handling, server-authoritative game logic, rate limiting
- AI/ML Integration:** Gemini, Groq (Llama 3), OpenCV, Tesseract OCR, PyTorch, Hugging Face, RAG pipelines, ChromaDB vector store
- DevOps & Tools:** Docker, Fly.io, Vercel, Git, Linux/Bash, Postman

PROJECTS

- Kallanum Policeum — Real-Time Multiplayer Social Deduction Game** *Node.js, React, Supabase Realtime, Ably, Agora Voice SDK | kallanumpoliceum.online (Live)*
 - Designed and built a full-stack browser-based multiplayer game with real-time game state synchronisation across concurrent lobby sessions using Supabase Realtime and Ably Pub/Sub, plus in-game spatial voice chat via Agora Voice SDK
 - Implemented the full game loop (role assignment, voting, elimination, win conditions) as server-authoritative logic with isolated per-lobby state, preventing race conditions and client-side manipulation across simultaneous sessions
 - Hardened the application against XSS, CSRF, and room enumeration attacks; implemented token-based room access, server-side rate limiting, and input sanitisation — shipped and live at kallanumpoliceum.online
- MedArchive — Full-Stack AI Medical Companion App** *Python, FastAPI, PostgreSQL, ChromaDB, Groq, Gemini, React/Vite, Supabase, Fly.io | GitHub*
 - Architected and shipped a full-stack product: React/Vite frontend, async FastAPI backend, Supabase auth with Row-Level Security, Telegram Bot interface, and Fly.io cloud deployment — covering the complete SDLC from design to production
 - Built a RAG pipeline with ChromaDB vector search and BAAI/bge embeddings for AI-powered Q&A over medical documents; integrated a dual-stage OCR pipeline (OpenCV + Tesseract, Gemini Flash fallback) for scanned prescriptions
 - Designed Pydantic-validated async endpoints for document ingestion, vector indexing, retrieval, and family health profile management; implemented background task queuing for long-running processing jobs
- CPMS — Campus Placement Management System** *Python, FastAPI, PostgreSQL, Supabase, Gemini API | GitHub*
 - Built a role-based REST API system in FastAPI serving 3 user roles (student, recruiter, admin) across the full recruitment lifecycle; designed a normalised PostgreSQL schema with foreign keys, joins, and Row-Level Security policies
 - Integrated Gemini 1.5 Flash to auto-score resumes against job descriptions using LLM-based semantic matching; implemented PDF parsing with structured field extraction, paginated endpoints, and JWT session management
- ACLInsight-Net — Deep Learning Diagnostic System** *Python, PyTorch, EfficientNet-B4, Grad-CAM, MRNet Dataset | GitHub*
 - Designed and implemented the full ML pipeline in Python — data loading, multi-plane MRI fusion, model training, evaluation, and inference — achieving AUC-ROC 0.9512 and 90.83% accuracy with 5-fold cross-validation; submitted to IEEE
 - Debugged and optimised the training pipeline: resolved CUDA OOM via gradient checkpointing and FP16 mixed-precision; built resumable epoch-level checkpointing for reliability on long training runs

OPEN SOURCE CONTRIBUTIONS

- OmniVoice TTS** — Contributed `ml_normalize.py` (Malayalam text normalization: numbers, currency, dates, units) to an open-source TTS library; delivered a FastAPI demo and Colab notebook — PR merged PR Merged

CERTIFICATIONS & ACHIEVEMENTS

- Complete 2026 Python Bootcamp: Learn Python from Scratch** *CodeWithHarry* 07/2026
- The Complete Full-Stack Web Development Bootcamp** *Udemy — Dr. Angela Yu* 06/2025
- Kaggle Hackathon — Avengers: Doomsday (Network Intrusion Classification)** *Ranked #5 · Score: 0.9824* 2025

LEADERSHIP & ACTIVITIES

- Technical Lead**, Xorir — IT Association, MEA Engineering College (2024–2026). Organised workshops and mentoring sessions for 40+ students; coordinated annual department technical events.
- Active Member**, MEA FOSS — Open-Source Club (2023–2026). Delivered sessions on backend architecture and API design; guided members through open-source contribution workflows.

ADDITIONAL INFORMATION

- Languages:** English (Professional), Malayalam (Native), Hindi (Conversational)